

SIMATS ENGINEERING
SAVEETHA INSTITUTE OF MEDICAL AND
TECHNICAL SCIENCES

CHENNAI – 602105

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 — Dashboard Development Task

Course Code: DSA0502

CO Assessed: CO5

Weightage: 25%

Course Title: Query Processing for Data Science

Assessment: Tool3–Dashboard Development Task

Total Marks:

CO5 Statement: Create meaningful visualizations using Pandas and Matplotlib, including charts, distributions, relationships, and interactive visual representations for effective communication

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Section Batch: / 2024

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Reg. No.: 192424422

Date of Submission: 24/08/26

Project Mode: Individual Submission

Code of Conduct

I Sandra Sudevan(192424422)(Name / Reg No) certify that this submission is my original work and that I have adhered to all guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature: Sandra Sudevan

1. Hospital Patient Analysis Dashboard

Build a dashboard using patient data containing **age, blood pressure,**

cholesterol, BMI, treatment cost, and admission date. Present distributions, counts, and relationships between health-related variables using Matplotlib. Include time-based plots to analyze admissions and use **lag/autocorrelation plots** to identify recurring admission patterns.

Aim

The aim of this project is to analyze hospital patient data and understand **patient health conditions, treatment costs, admission trends, and relationships between medical variables** using Python and Matplotlib.

Dataset

The dataset contains:

- **Age** – age of the patient
- **Blood Pressure** – patient's blood pressure
- **Cholesterol** – cholesterol level
- **BMI** – Body Mass Index
- **Treatment Cost** – cost of treatment
- **Admission Date** – date on which the patient was admitted

Step-by-step Explanation

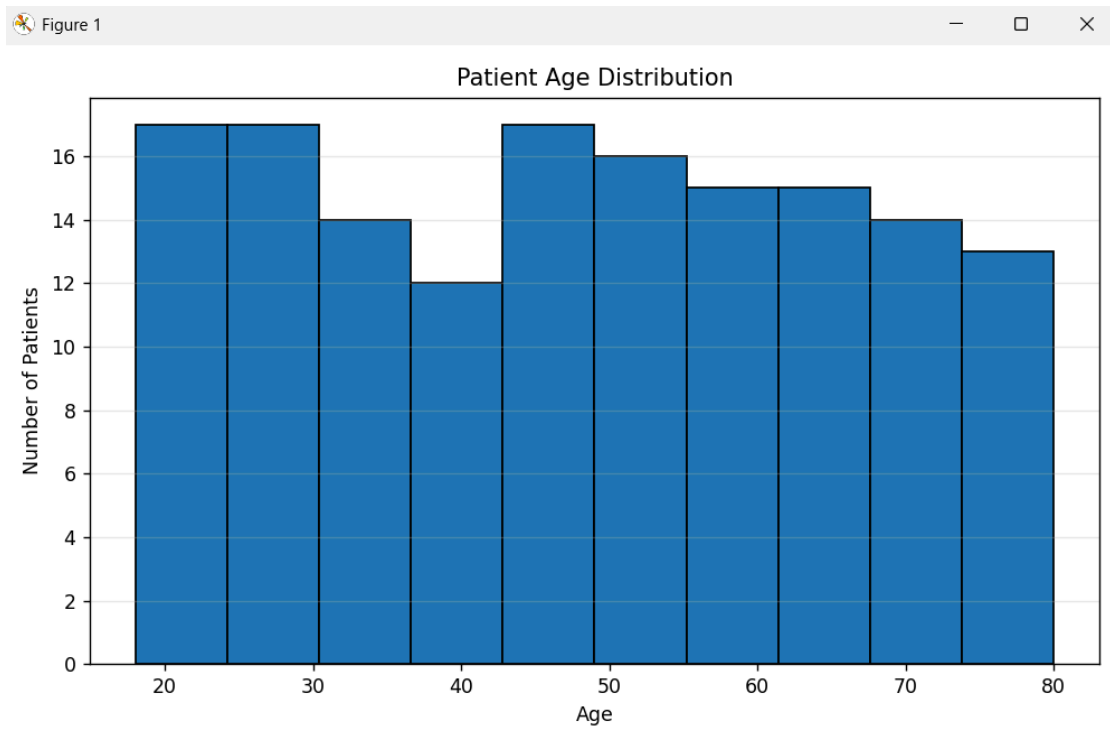
1. Create/Load the dataset

The patient information is stored in a Pandas DataFrame. If an actual CSV file is available, it can also be loaded using `pd.read_csv()`.

2. Age Distribution

A histogram is used to show how many patients belong to different age groups.

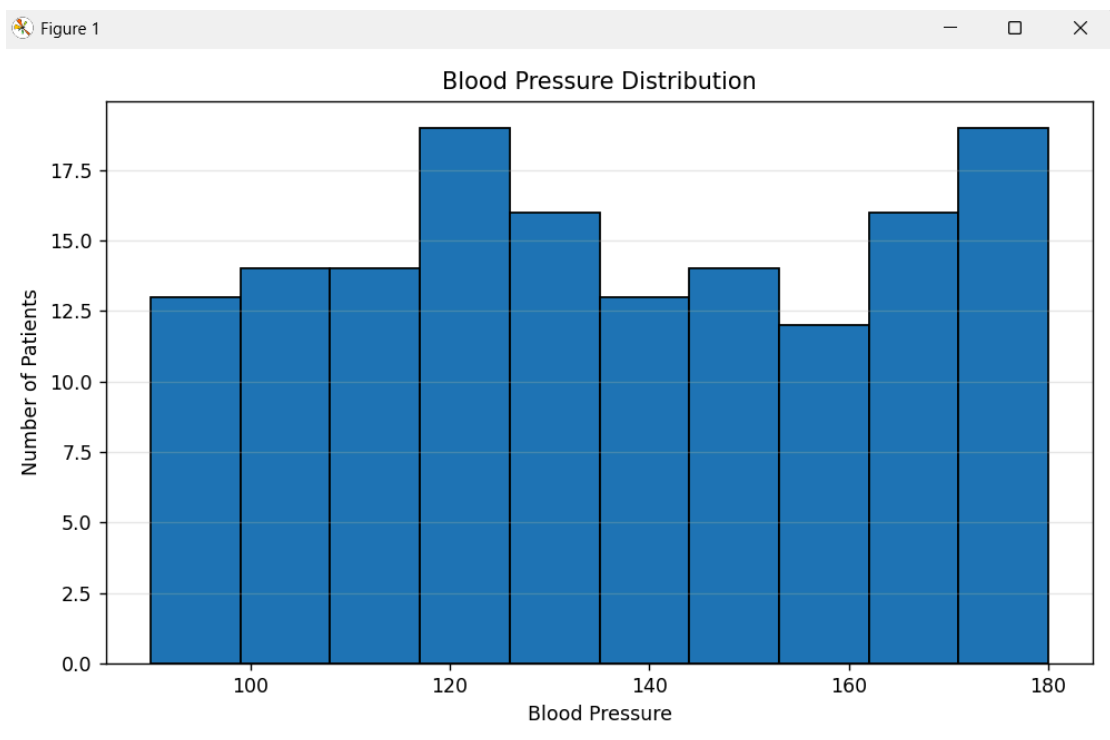
It helps identify whether most patients are young, middle-aged, or elderly.



3. Blood Pressure Distribution

A histogram displays the distribution of blood-pressure values among patients.

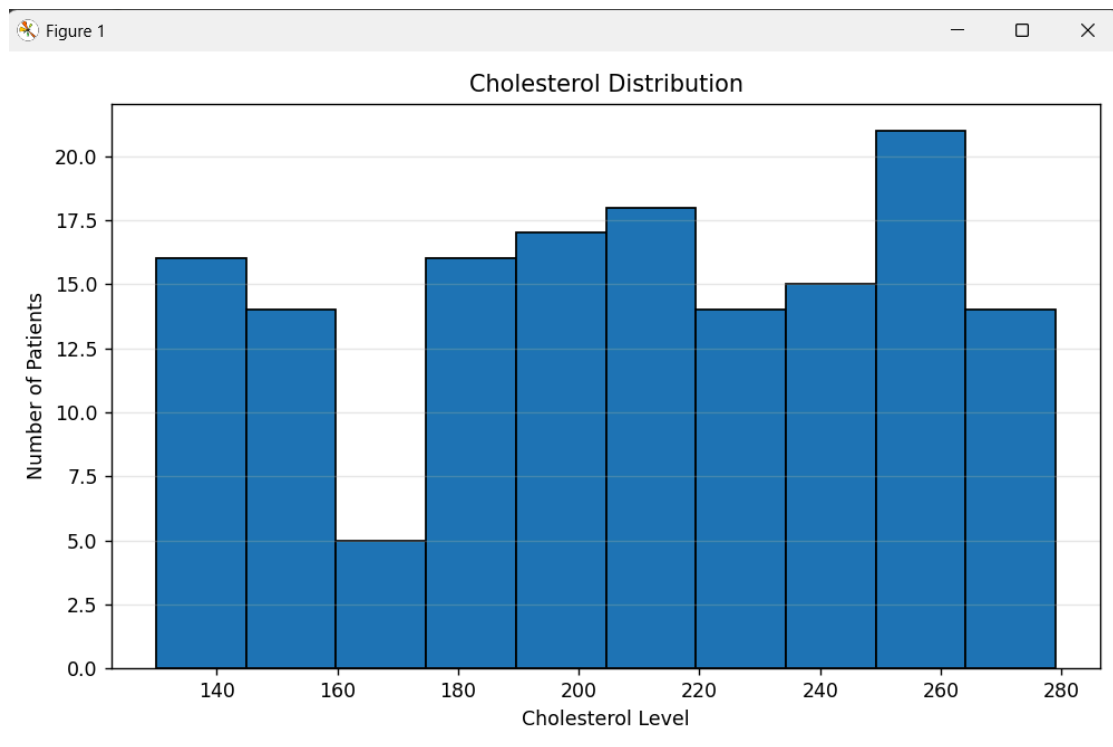
This helps identify whether the patients generally have low, normal, or high blood pressure.



4. Cholesterol Distribution

The cholesterol values are plotted using a histogram.

It helps understand the range and frequency of cholesterol levels.



5. BMI Distribution

The BMI values are displayed using a histogram.

This helps identify whether patients generally have normal, overweight, or higher BMI values.

6. Treatment Cost Distribution

A histogram is used to understand the distribution of hospital treatment costs.

It helps identify whether most treatments are inexpensive or expensive.

7. Relationship Between Age and Blood Pressure

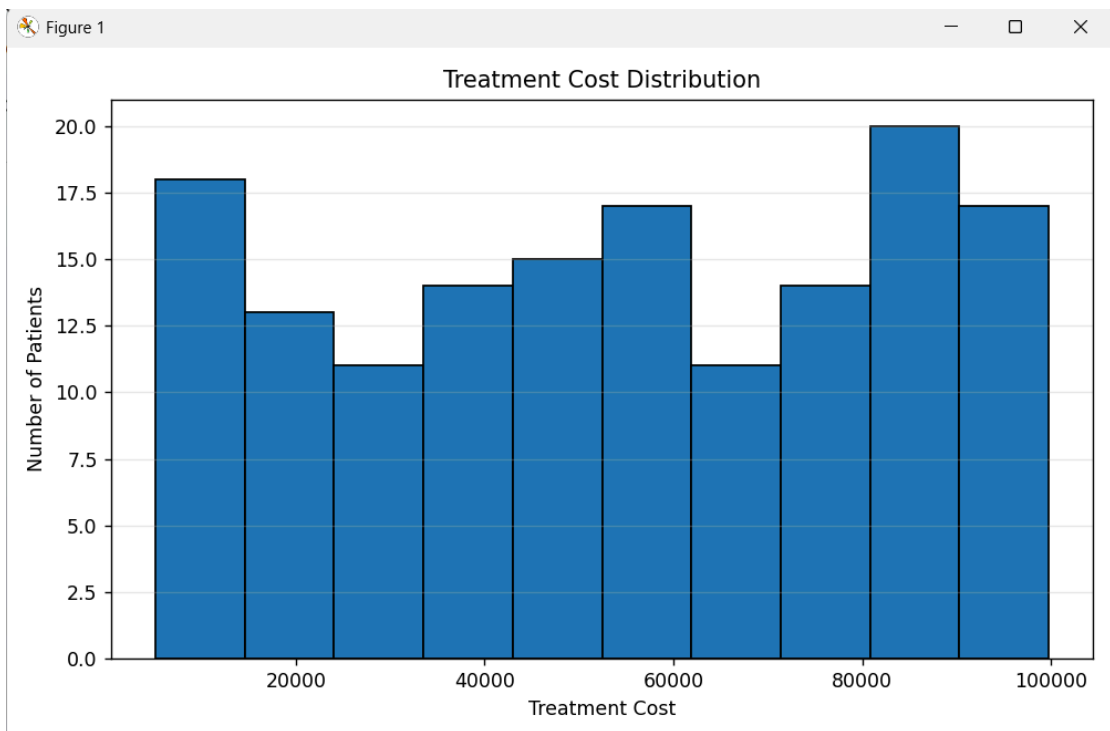
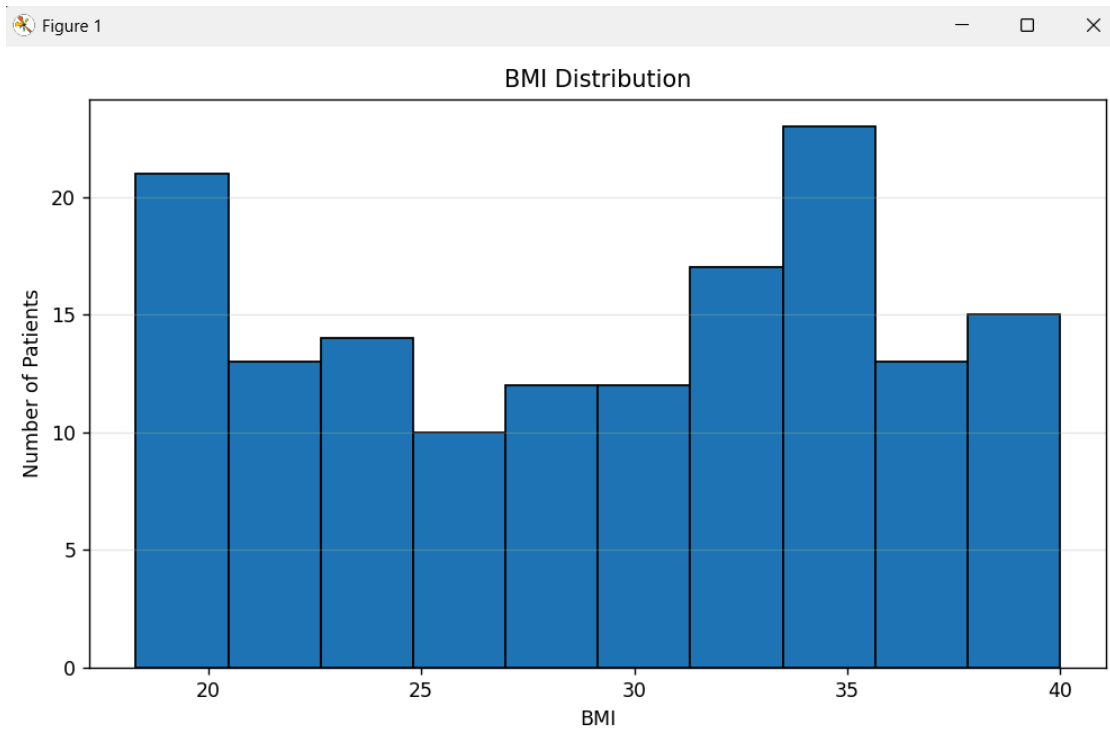
A scatter plot is used to compare age with blood pressure.

It helps determine whether blood pressure tends to increase or decrease with age.

8. BMI and Treatment Cost

A scatter plot compares BMI with treatment cost.

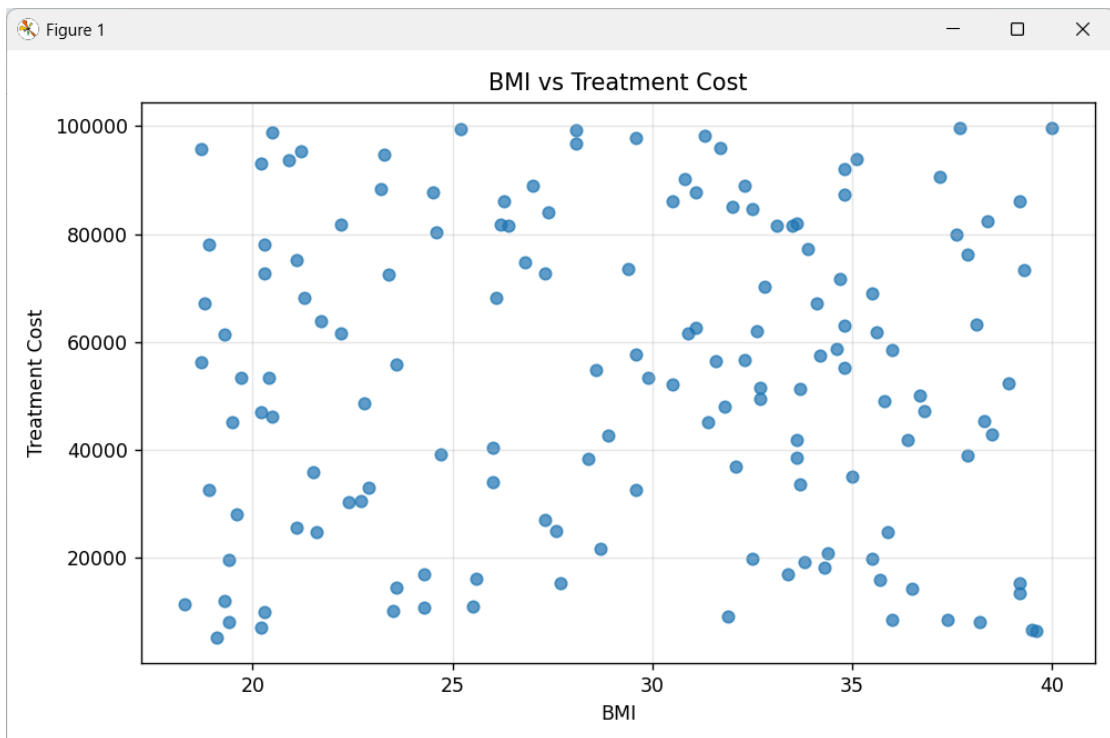
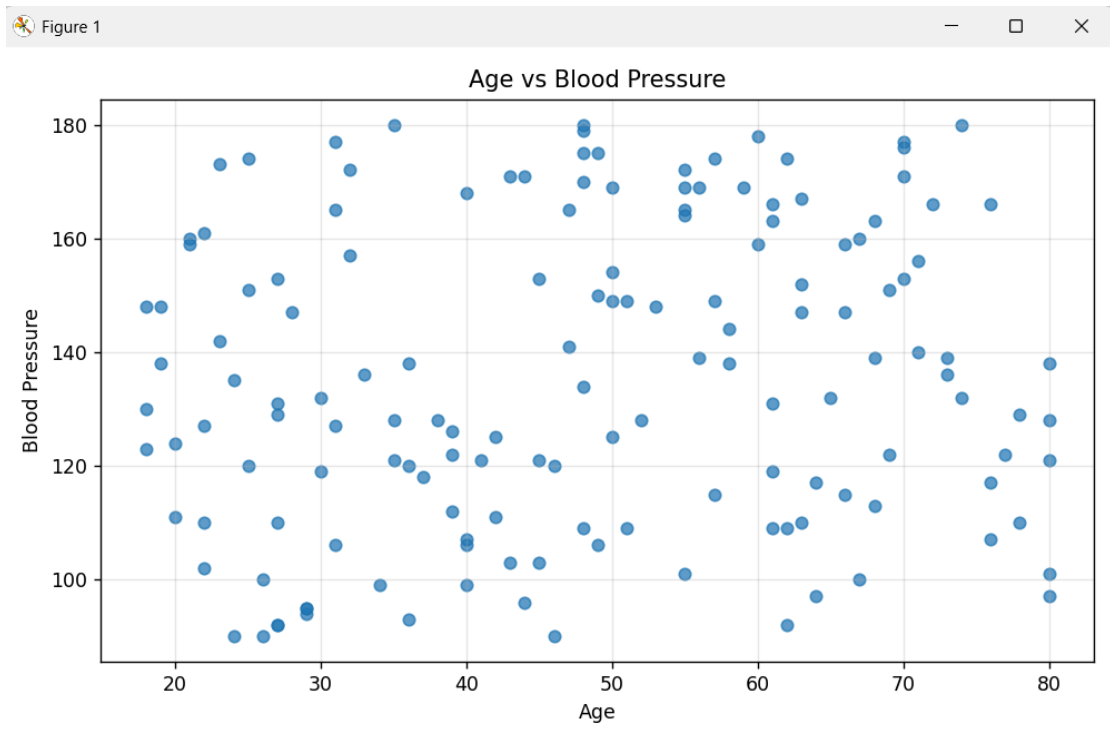
It helps investigate whether patients with higher BMI tend to have higher treatment costs.



9. Age and Cholesterol

A scatter plot compares age and cholesterol.

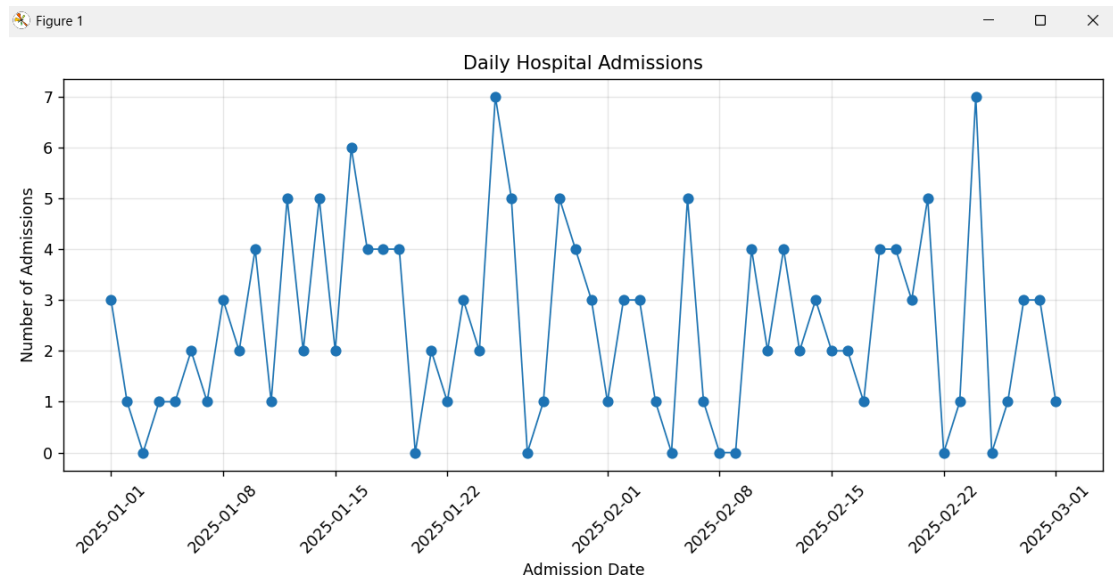
It helps identify possible relationships between increasing age and cholesterol levels.



10. Admission Trend

The admission dates are grouped by day to calculate the number of patients admitted each day.

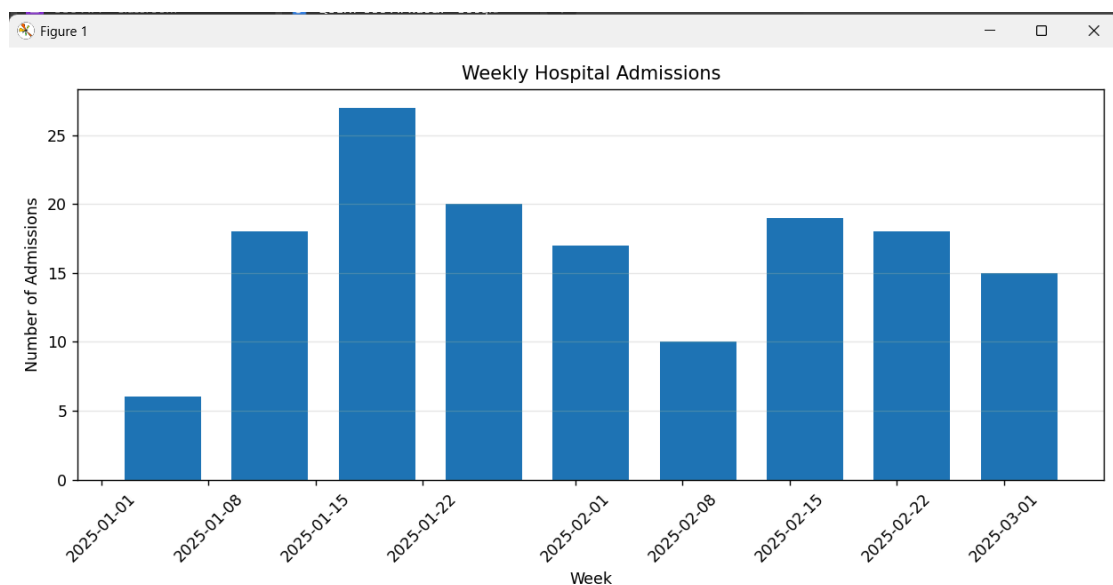
A line graph shows how hospital admissions change over time.



11. Weekly Admissions

Daily admissions are resampled into weekly totals.

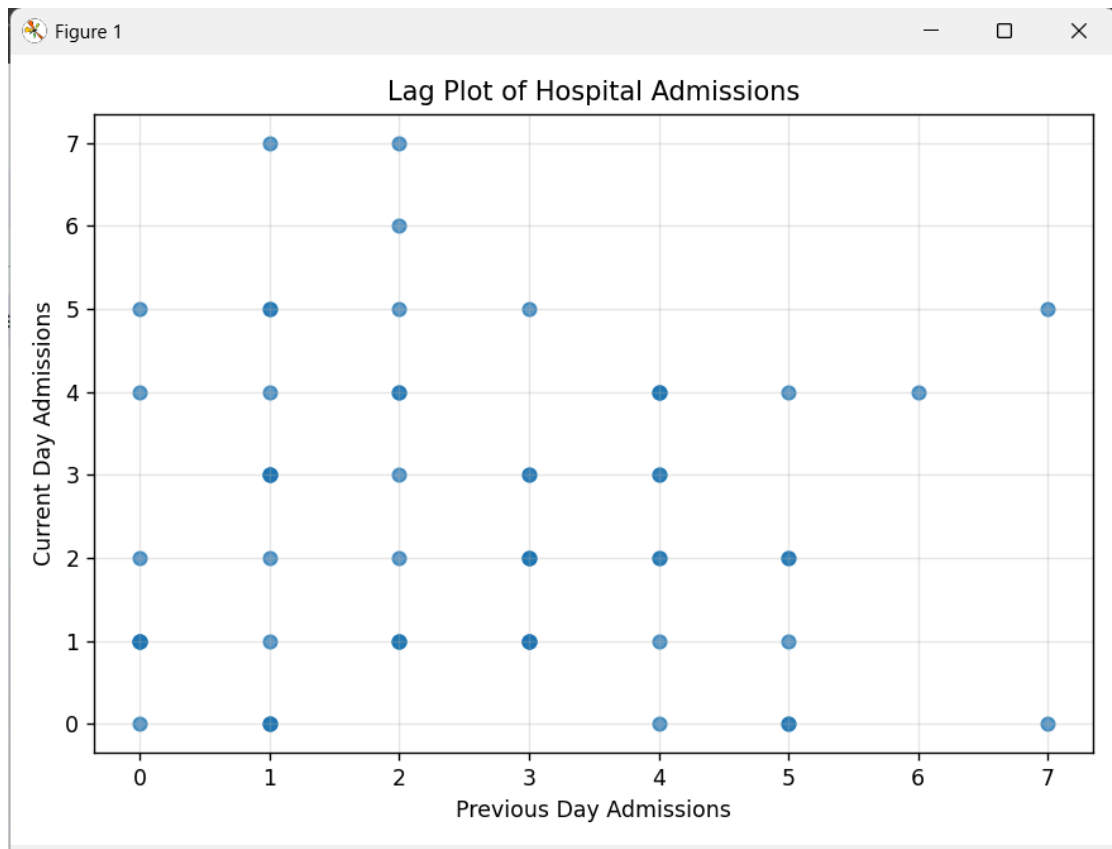
A bar graph makes it easier to compare the number of admissions from week to week.



12. Lag Plot

A lag plot compares the current day's admission count with the previous day's admission count.

If the points show a pattern, it indicates that today's admissions may be related to previous admissions.



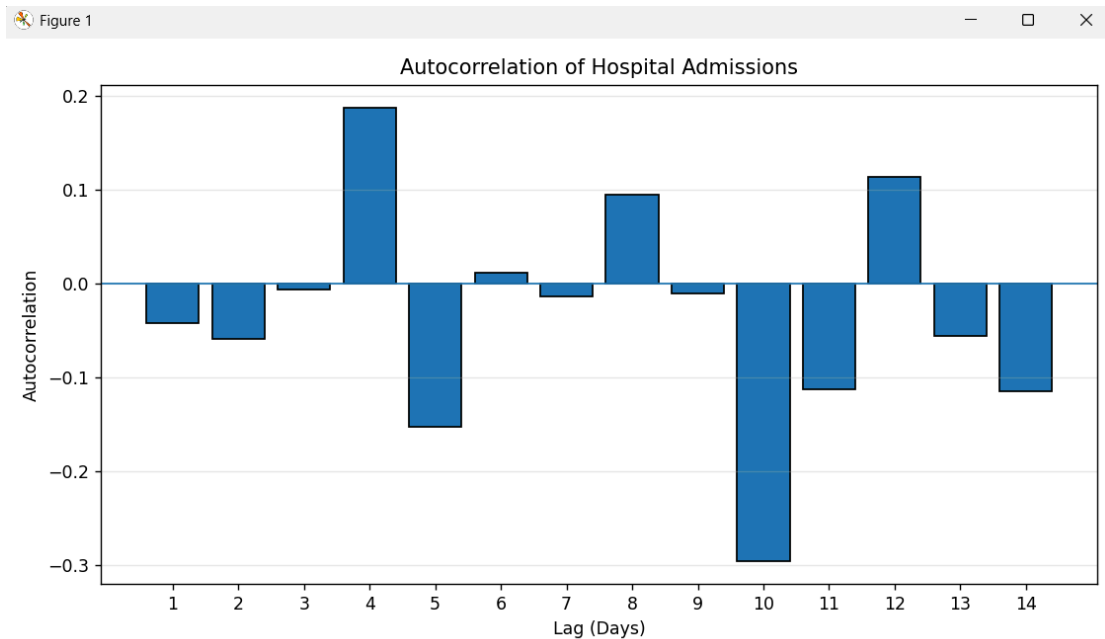
13. Autocorrelation

Autocorrelation measures the relationship between admission counts at different time lags.

For example:

- Lag 1 → today's admissions vs yesterday's
- Lag 2 → today's admissions vs admissions two days ago
- Lag 7 → today's admissions vs admissions seven days ago

A high value at a particular lag can indicate a **repeating admission pattern**.



Conclusion

The Hospital Patient Analysis Dashboard helps understand **patient characteristics, health-variable relationships, treatment costs, and admission patterns**. Matplotlib provides different visualizations, while lag and autocorrelation analysis help identify patterns in hospital admissions.

2. Stock Market Pattern Analysis Dashboard

Develop a dashboard using historical stock data containing **date, opening price, closing price, high, low, and trading volume**. Visualize price evolution over time, distributions, and relationships between price and volume. Use **Pandas plotting, scatter plots, lag plots, autocorrelation plots, and bootstrap plots** to investigate trends, dependencies, and uncertainty in stock behavior.

The **Stock Market Pattern Analysis Dashboard** is used to analyze historical stock-market data and identify trends, relationships, dependencies, and uncertainty in stock prices.

The dataset contains **Date, Opening Price, Closing Price, High Price, Low Price, and Trading Volume**.

First, the stock prices are analyzed over time using line graphs. This helps identify whether the stock price is increasing, decreasing, or fluctuating. The high and low prices show the daily price range and help understand price variation.

Histograms are used to study the distribution of **closing prices and trading volume**. These graphs show which price and volume ranges occur most frequently.

Scatter plots are used to study relationships between variables. For example, **closing price vs trading volume** helps determine whether trading activity is related to stock

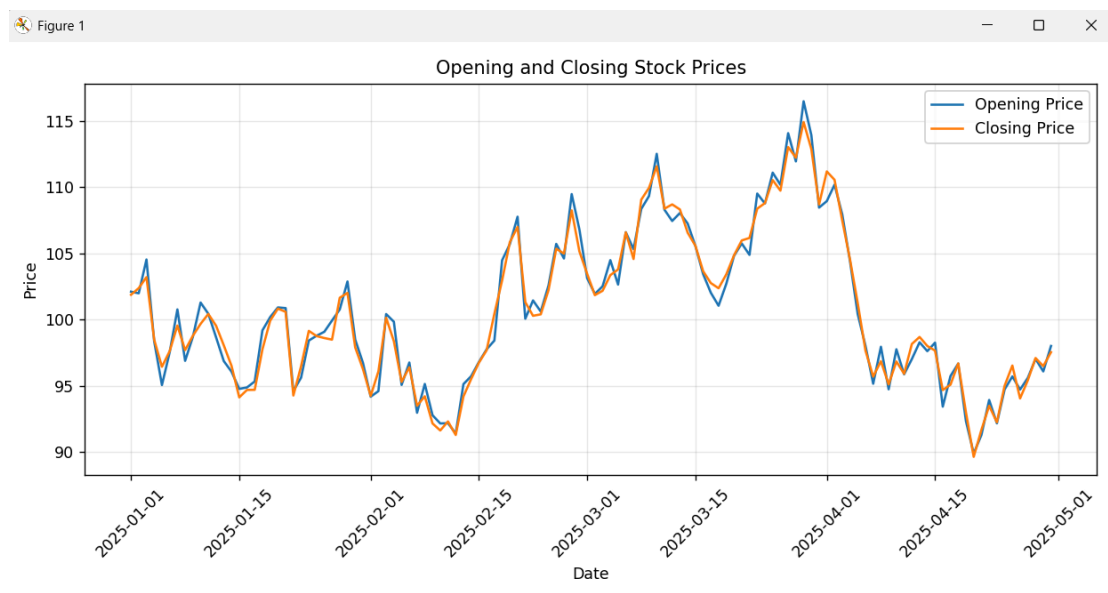
price. **Opening price vs closing price** shows how strongly the opening and closing prices are related.

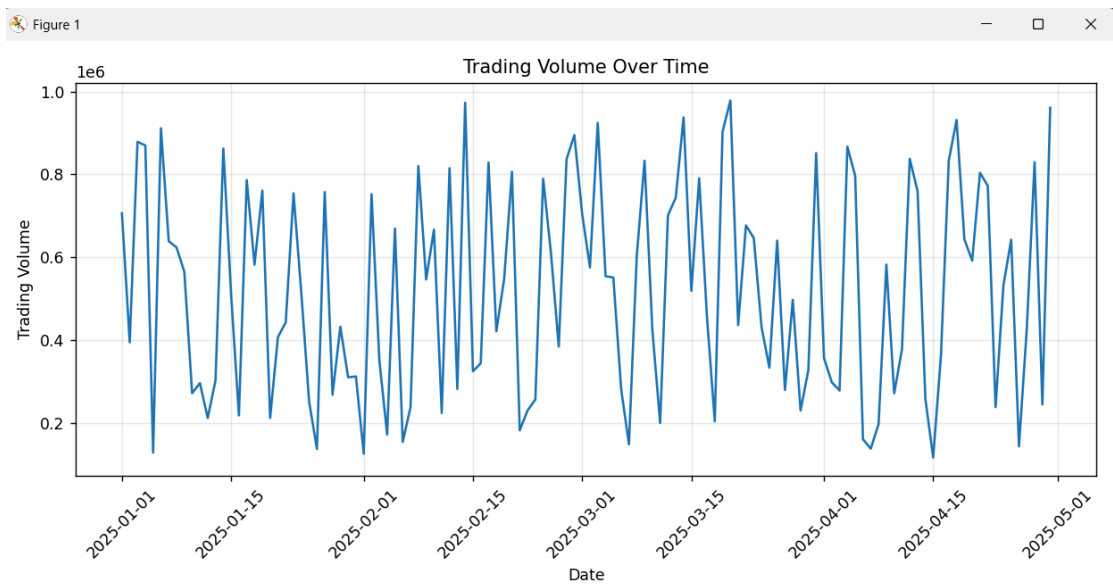
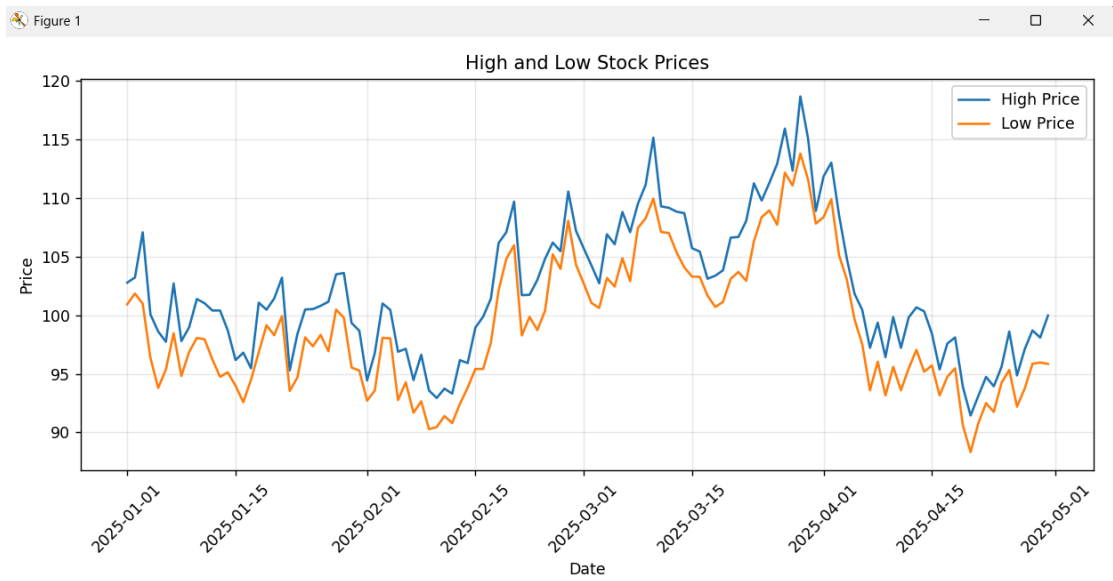
Daily stock returns are calculated to measure the percentage change in the closing price from one day to the next. The return distribution helps understand whether the stock generally produces positive, negative, or small returns.

A **lag plot** compares the current day's return with the previous day's return. It helps identify whether the current stock movement depends on the previous day's movement.

Autocorrelation analysis measures the relationship between stock returns at different time lags. It helps identify whether there are recurring or dependent patterns in stock returns.

Finally, **bootstrap analysis** is used to estimate the uncertainty of the average stock return. The data is repeatedly resampled, and the mean return is calculated for each sample. A **95% confidence interval** is then obtained to understand the possible range of the average return.





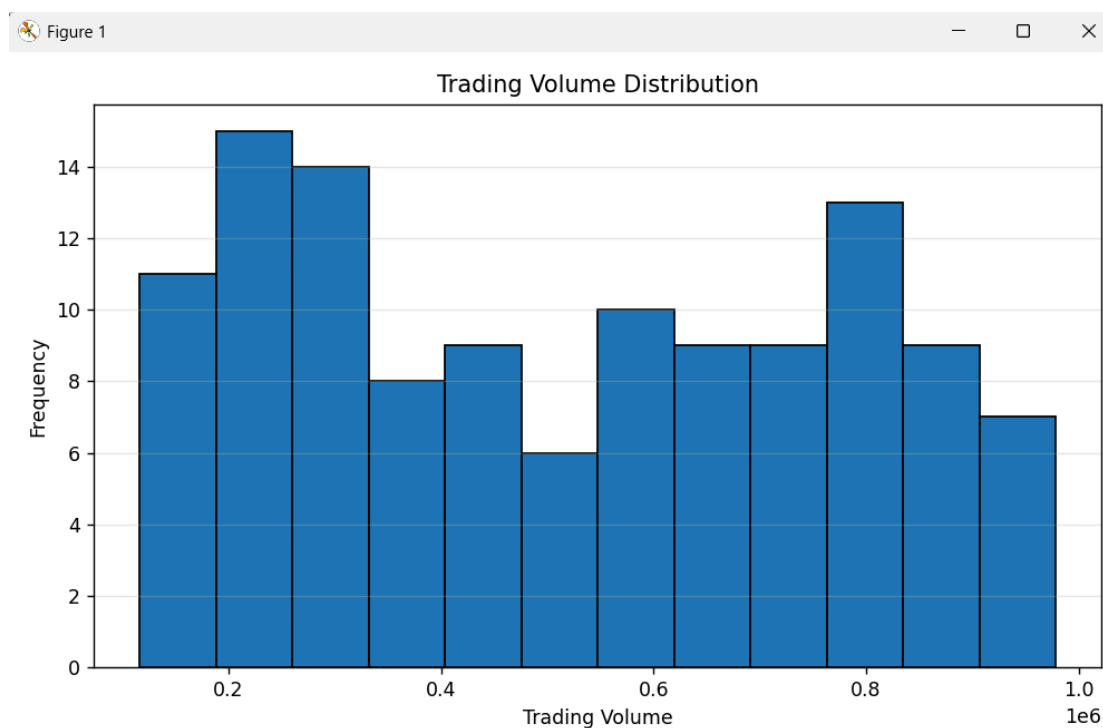
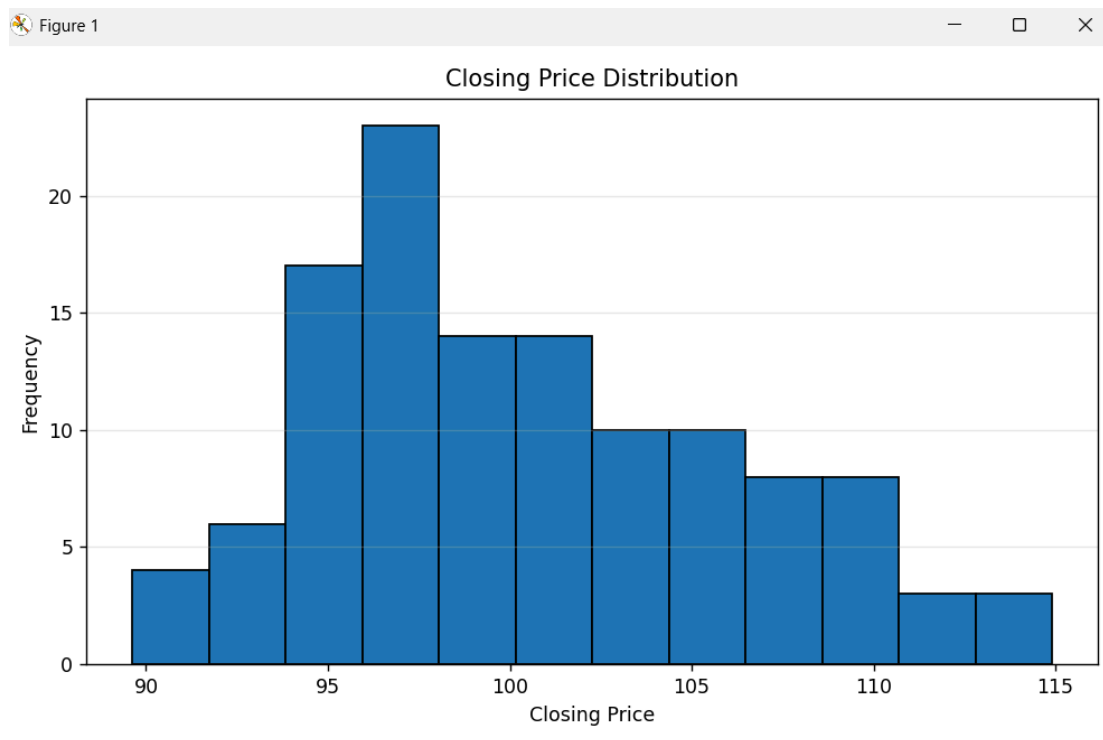




Figure 1

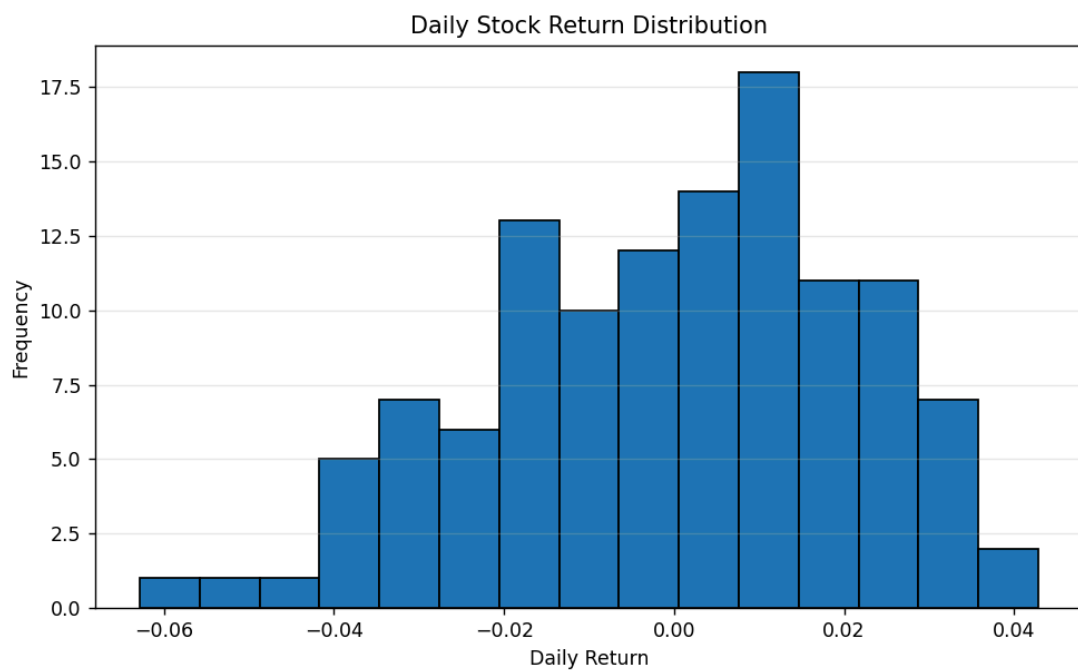
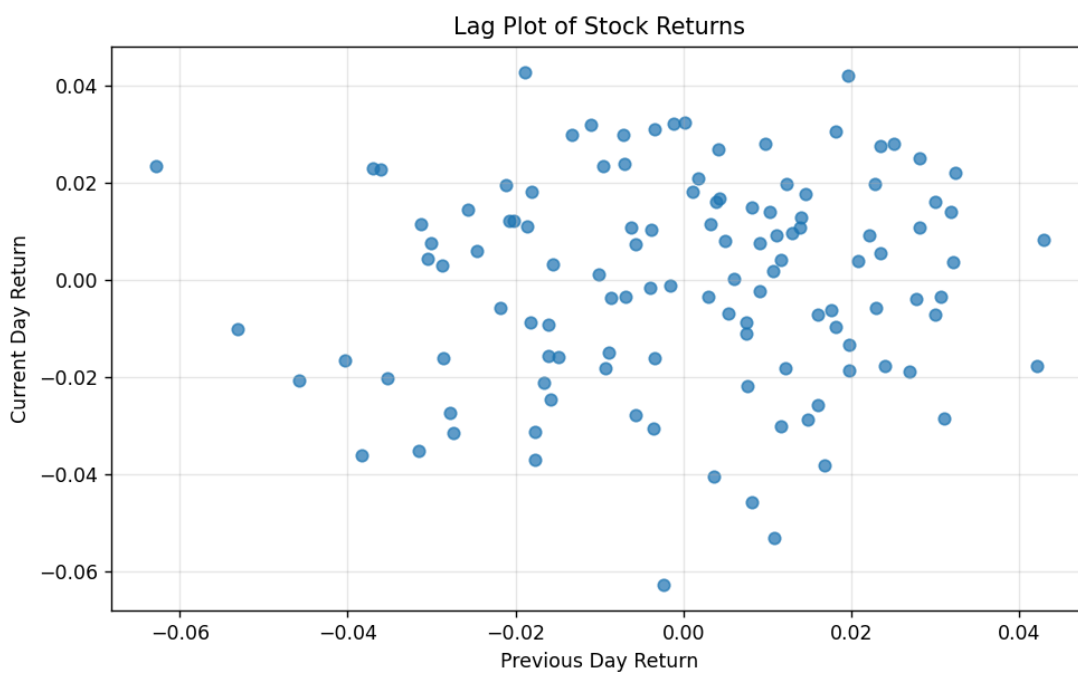
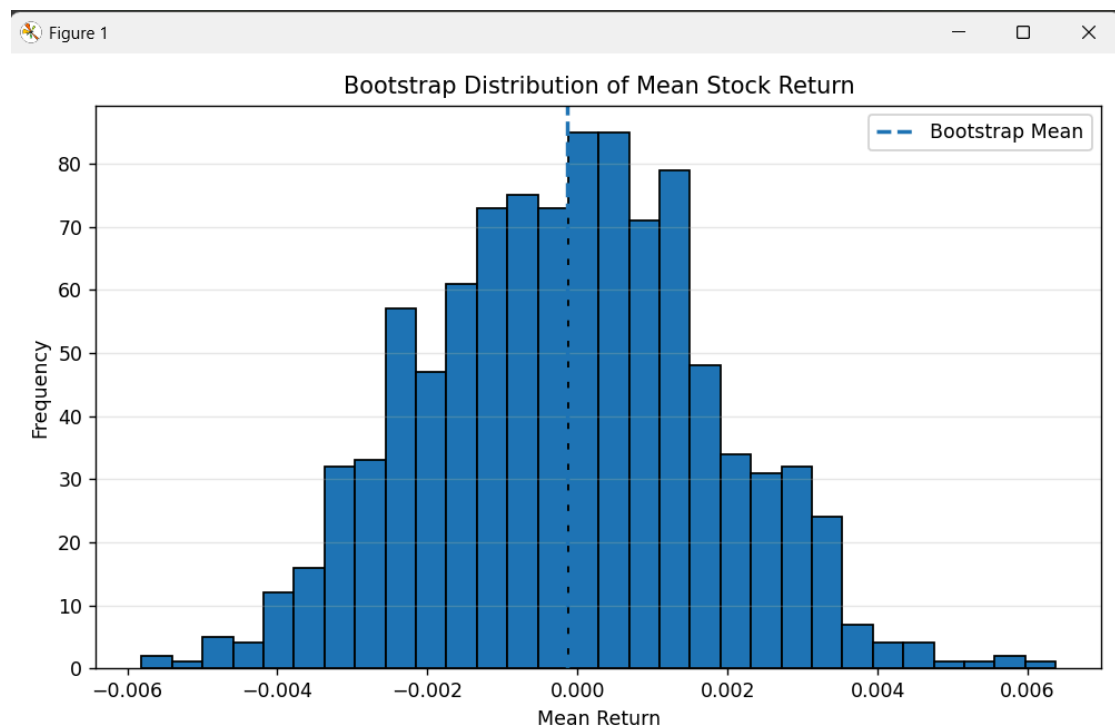
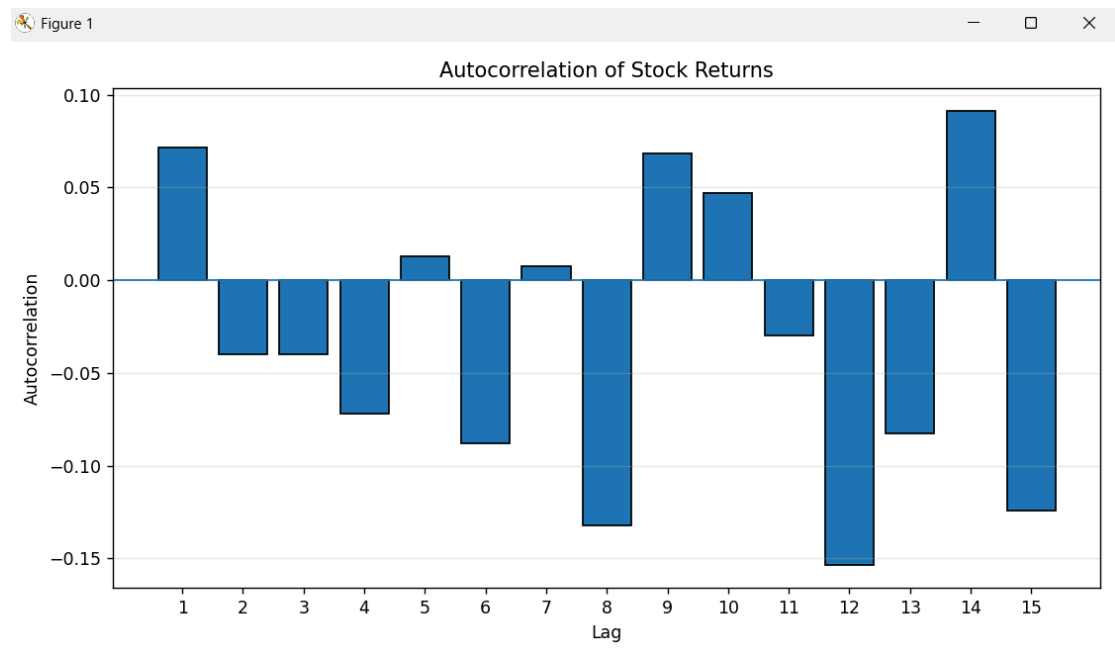


Figure 1





Conclusion

The dashboard provides a complete analysis of stock-market behavior by studying **price trends, trading volume, distributions, relationships, time dependencies, and uncertainty**. It helps understand how stock prices behave over time and whether previous market movements influence future returns.